

In [2]:

Dataset Loaded Successfully!

First 5 Rows:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	\
0	10/03/2004	18.00.00	2.6	1360.0	150.0	11.9	
1	10/03/2004	19.00.00	2.0	1292.0	112.0	9.4	
2	10/03/2004	20.00.00	2.2	1402.0	88.0	9.0	
3	10/03/2004	21.00.00	2.2	1376.0	80.0	9.2	
4	10/03/2004	22.00.00	1.6	1272.0	51.0	6.5	

	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	N02(GT)	PT08.S4(N02)	PT08.S5(03)	\
0	1046.0	166.0	1056.0	113.0	1692.0	1268.0	
1	955.0	103.0	1174.0	92.0	1559.0	972.0	
2	939.0	131.0	1140.0	114.0	1555.0	1074.0	
3	948.0	172.0	1092.0	122.0	1584.0	1203.0	
4	836.0	131.0	1205.0	116.0	1490.0	1110.0	

	T	RH	AH	Unnamed: 15	Unnamed: 16
0	13.6	48.9	0.7578	NaN	NaN
1	13.3	47.7	0.7255	NaN	NaN
2	11.9	54.0	0.7502	NaN	NaN
3	11.0	60.0	0.7867	NaN	NaN
4	11.2	59.6	0.7888	NaN	NaN

Dataset Shape:
(9471, 17)

Column Names:

```
Index(['Date', 'Time', 'CO(GT)', 'PT08.S1(CO)', 'NMHC(GT)', 'C6H6(GT)',
      'PT08.S2(NMHC)', 'NOx(GT)', 'PT08.S3(NOx)', 'N02(GT)', 'PT08.S4(N02)',
      'PT08.S5(03)', 'T', 'RH', 'AH', 'Unnamed: 15', 'Unnamed: 16'],
      dtype='object')
```

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 9471 entries, 0 to 9470

Data columns (total 17 columns):

#	Column	Non-Null Count	Dtype
0	Date	9357 non-null	object
1	Time	9357 non-null	object
2	CO(GT)	9357 non-null	float64
3	PT08.S1(CO)	9357 non-null	float64
4	NMHC(GT)	9357 non-null	float64

```

5   C6H6(GT)          9357 non-null    float64
6   PT08.S2(NMHC)     9357 non-null    float64
7   NOx(GT)           9357 non-null    float64
8   PT08.S3(NOx)      9357 non-null    float64
9   NO2(GT)           9357 non-null    float64
10  PT08.S4(NO2)      9357 non-null    float64
11  PT08.S5(O3)       9357 non-null    float64
12  T                 9357 non-null    float64
13  RH                9357 non-null    float64
14  AH                9357 non-null    float64
15  Unnamed: 15       0 non-null       float64
16  Unnamed: 16       0 non-null       float64
dtypes: float64(15), object(2)
memory usage: 1.2+ MB
None

```

Dataset Shape After Removing Empty Columns:
(9471, 15)

Missing Values Before Cleaning:

```

Date          114
Time          114
CO(GT)        114
PT08.S1(CO)   114
NMHC(GT)      114
C6H6(GT)      114
PT08.S2(NMHC) 114
NOx(GT)       114
PT08.S3(NOx)  114
NO2(GT)       114
PT08.S4(NO2)  114
PT08.S5(O3)   114
T             114
RH            114
AH            114
dtype: int64

```

Missing Values After Replacing -200:

```

Date          114
Time          114
CO(GT)        1797
PT08.S1(CO)    480
NMHC(GT)      8557
C6H6(GT)      480
PT08.S2(NMHC)  480
NOx(GT)       1753
PT08.S3(NOx)   480
NO2(GT)       1756
PT08.S4(NO2)   480
PT08.S5(O3)    480
T             480
RH            480
AH            480
dtype: int64

```

Duplicate Rows: 113

Dataset Shape After Removing Duplicates:
(9358, 15)

Missing Values After Filling:

```
Date          1
Time          1
CO(GT)        0
PT08.S1(CO)   0
NMHC(GT)      0
C6H6(GT)      0
PT08.S2(NMHC) 0
NOx(GT)       0
PT08.S3(NOx)  0
NO2(GT)       0
PT08.S4(NO2)  0
PT08.S5(O3)   0
T             0
RH            0
AH            0
```

dtype: int64

Original Dataset Shape : (9358, 15)
After Removing Outliers : (7175, 15)

First 5 Rows After Scaling:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)
\						
0	10/03/2004	18.00.00	0.978136	1.890609	0.0	0.619859
23	11/03/2004	17.00.00	1.330075	2.031972	0.0	0.488720
39	12/03/2004	09.00.00	0.039633	3.027660	0.0	2.530743
94	14/03/2004	16.00.00	1.447388	2.492939	0.0	0.619859
184	18/03/2004	10.00.00	3.207082	3.470188	0.0	2.380869

	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)
\					
0	0.717249	-0.231132	1.172337	0.209656	0.870464
23	0.593714	0.449406	0.918121	0.845034	0.956406
39	2.175909	-0.107398	-0.358252	0.094133	2.035459
94	0.712497	-0.195780	0.738052	0.671749	0.724043
184	2.080882	1.386251	-0.787240	1.278246	1.882673

	PT08.S5(O3)	T	RH	AH
0	1.050615	-0.636929	0.038427	-0.753993
23	0.510392	-1.059978	1.123951	-0.605359
39	2.104709	-1.126775	0.462187	-1.003023
94	0.493922	0.320498	-1.151586	-0.769175
184	2.865633	-0.169348	-0.449188	-0.599237

Final Dataset Information:

<class 'pandas.core.frame.DataFrame'>

Index: 7175 entries, 0 to 9357

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	Date	7174 non-null	object
1	Time	7174 non-null	object

```

2  CO(GT)          7175 non-null float64
3  PT08.S1(CO)     7175 non-null float64
4  NMHC(GT)        7175 non-null float64
5  C6H6(GT)        7175 non-null float64
6  PT08.S2(NMHC)   7175 non-null float64
7  NOx(GT)         7175 non-null float64
8  PT08.S3(NOx)    7175 non-null float64
9  NO2(GT)         7175 non-null float64
10 PT08.S4(NO2)    7175 non-null float64
11 PT08.S5(O3)     7175 non-null float64
12 T              7175 non-null float64
13 RH             7175 non-null float64
14 AH             7175 non-null float64

```

dtypes: float64(13), object(2)

memory usage: 896.9+ KB

None

Statistical Summary:

	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC) \
count	7.175000e+03	7.175000e+03	7175.0	7.175000e+03	7.175000e+03
mean	-2.337116e-16	-4.080050e-16	0.0	1.901383e-16	-1.901383e-16
std	1.000070e+00	1.000070e+00	0.0	1.000070e+00	1.000070e+00
min	-1.954687e+00	-2.368723e+00	0.0	-1.515835e+00	-2.162061e+00
25%	-6.642445e-01	-7.584131e-01	0.0	-7.852025e-01	-7.699198e-01
50%	3.963296e-02	-8.232855e-02	0.0	-1.669755e-01	-3.821399e-02
75%	3.915717e-01	6.121946e-01	0.0	5.636565e-01	6.721108e-01
max	3.441707e+00	3.568528e+00	0.0	3.467450e+00	2.788831e+00

	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3) \
count	7.175000e+03	7.175000e+03	7.175000e+03	7.175000e+03	7.175000e+03
mean	9.506913e-17	2.535177e-16	1.584486e-17	3.802765e-16	1.188364e-16
std	1.000070e+00	1.000070e+00	1.000070e+00	1.000070e+00	1.000070e+00
min	-1.609886e+00	-2.169536e+00	-2.591780e+00	-2.538581e+00	-2.177548e+00
25%	-7.172313e-01	-7.130941e-01	-6.278865e-01	-6.955960e-01	-7.396979e-01
50%	-1.073981e-01	-1.411095e-01	9.413326e-02	1.224473e-01	-5.947781e-02
75%	4.052153e-01	6.109444e-01	4.984643e-01	6.858468e-01	6.421536e-01
max	3.092017e+00	2.999510e+00	2.808927e+00	2.751645e+00	3.531030e+00

	T	RH	AH
count	7.175000e+03	7.175000e+03	7.175000e+03
mean	1.267588e-16	3.802765e-16	-3.168971e-16
std	1.000070e+00	1.000070e+00	1.000070e+00
min	-2.362524e+00	-2.266135e+00	-2.153655e+00
25%	-7.037263e-01	-7.684603e-01	-6.642491e-01
50%	-1.348823e-02	4.423164e-02	-6.346694e-02
75%	6.767499e-01	7.466297e-01	7.220677e-01
max	2.547072e+00	2.348794e+00	2.580366e+00

```
=====
Data Cleaning Completed Successfully!
Cleaned Dataset Saved As:
AirQuality_Cleaned.csv
=====
```

```
/var/folders/2c/qjv1tmms0f31gbgrv83fydm40000gn/T/ipykernel_50287/109
9910489.py:57: FutureWarning: A value is trying to be set on a copy
of a DataFrame or Series through chained assignment using an inplace
method.
```

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df[col].fillna(median, inplace=True)
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df[col].fillna(median, inplace=True)
/var/folders/2c/qjv1tmms0f31gbgrv83fydm40000gn/T/ipykernel_50287/1099910489.py:85: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df_clean[numeric_columns] = scaler.fit_transform(df_clean[numeric_columns])
```

In [6]:

	Minimum Value	Maximum Value	Mean (Actual Value)	
Median				
C0(GT)	0.1000	11.900	2.089271	
1.8000				
PT08.S1(C0)	647.0000	2040.000	1098.388651	10
63.0000				
NMHC(GT)	7.0000	1189.000	156.720881	1
50.0000				
C6H6(GT)	0.1000	63.700	10.009254	
8.2000				
PT08.S2(NMHC)	383.0000	2214.000	937.970827	9
09.0000				
NOx(GT)	2.0000	1479.000	235.173007	1
80.0000				
PT08.S3(NOx)	322.0000	2683.000	834.336931	8
06.0000				
N02(GT)	2.0000	340.000	112.372943	1
09.0000				
PT08.S4(N02)	551.0000	2775.000	1456.528745	14
63.0000				
PT08.S5(03)	221.0000	2523.000	1020.556743	9
63.0000				
T	-1.9000	44.600	18.297521	
17.8000				
RH	9.2000	88.700	49.248547	
49.6000				
AH	0.1847	2.231	1.024349	
0.9954				

In [13]:

```

Date      1
Time      1
C0(GT)    0
PT08.S1(C0) 0
NMHC(GT)  0
C6H6(GT)  0
PT08.S2(NMHC) 0
NOx(GT)   0
PT08.S3(NOx) 0
N02(GT)   0
PT08.S4(N02) 0
PT08.S5(03) 0
T          0
RH         0
AH         0
dtype: int64

```

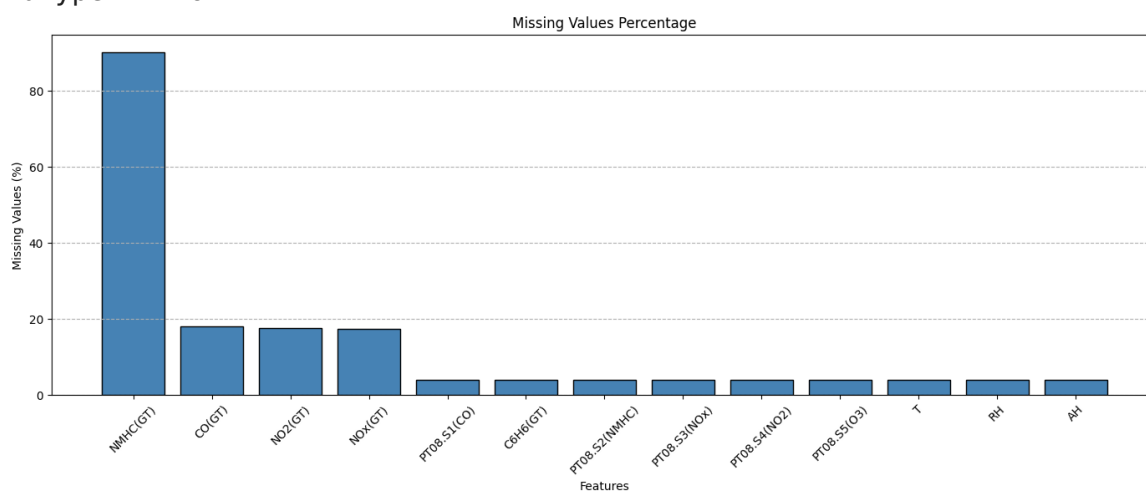
In [17]:

Shape: (9357, 15)

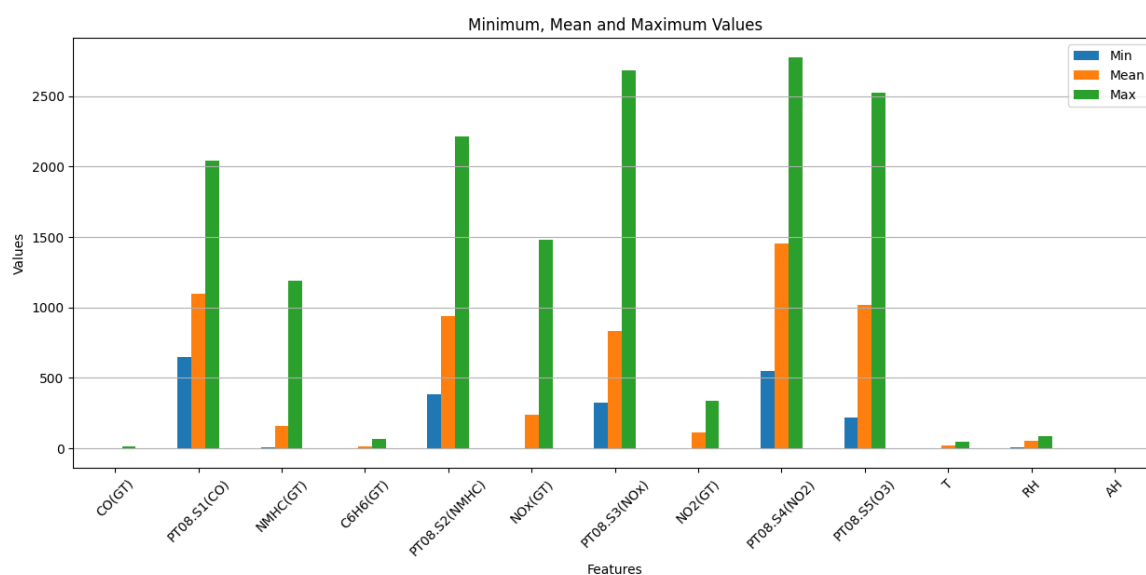
Missing Values:

Date	0
Time	0
CO(GT)	1683
PT08.S1(CO)	366
NMHC(GT)	8443
C6H6(GT)	366
PT08.S2(NMHC)	366
NOx(GT)	1639
PT08.S3(NOx)	366
NO2(GT)	1642
PT08.S4(NO2)	366
PT08.S5(O3)	366
T	366
RH	366
AH	366

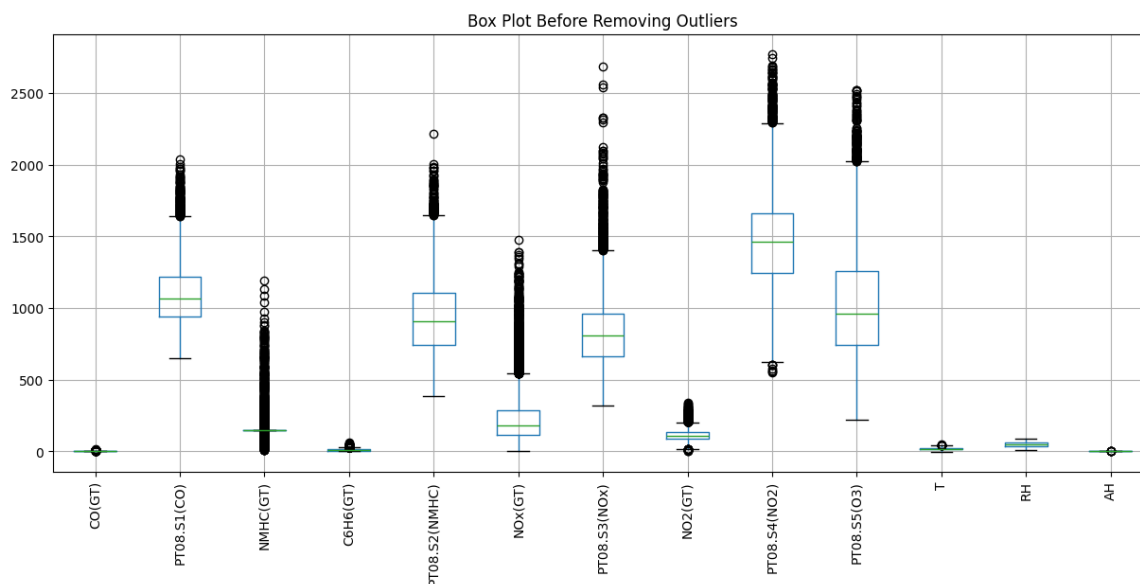
dtype: int64



In [8]:

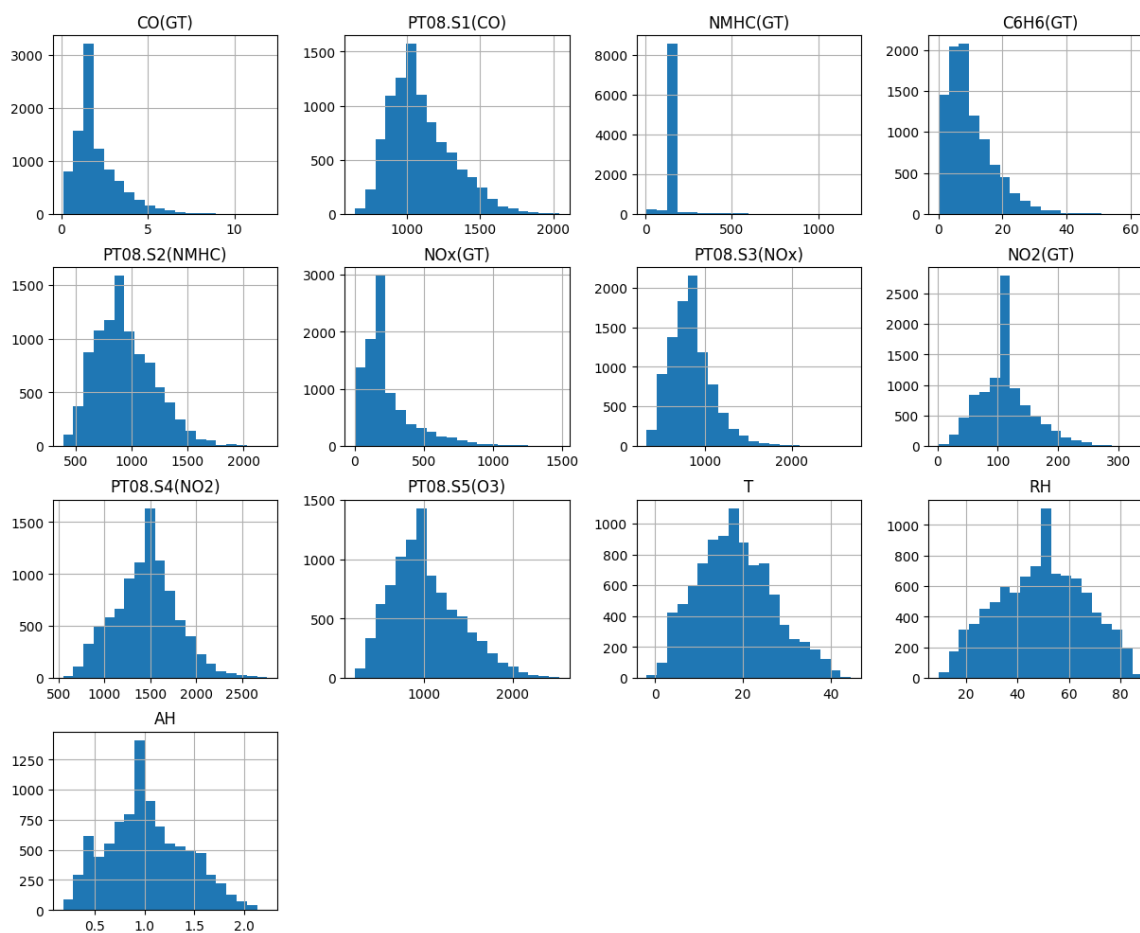


In [9]:

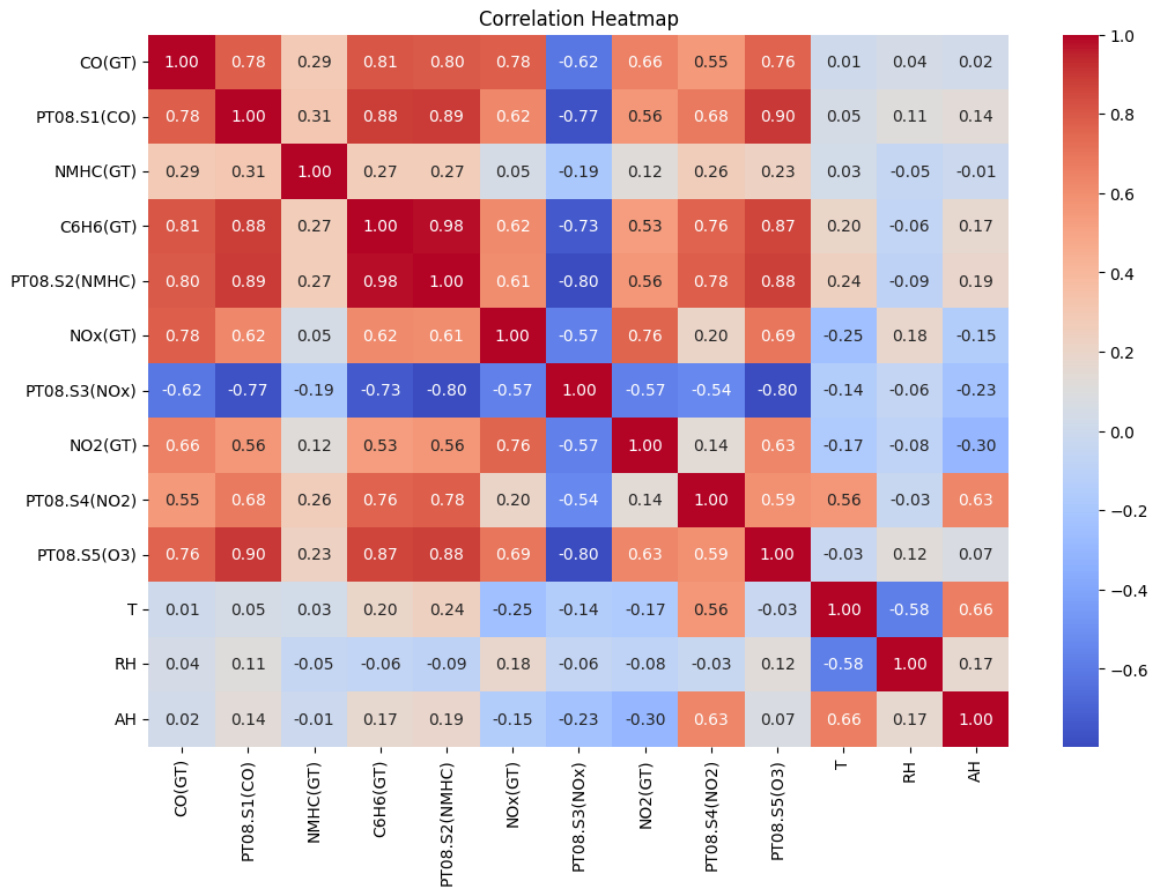


In [10]:

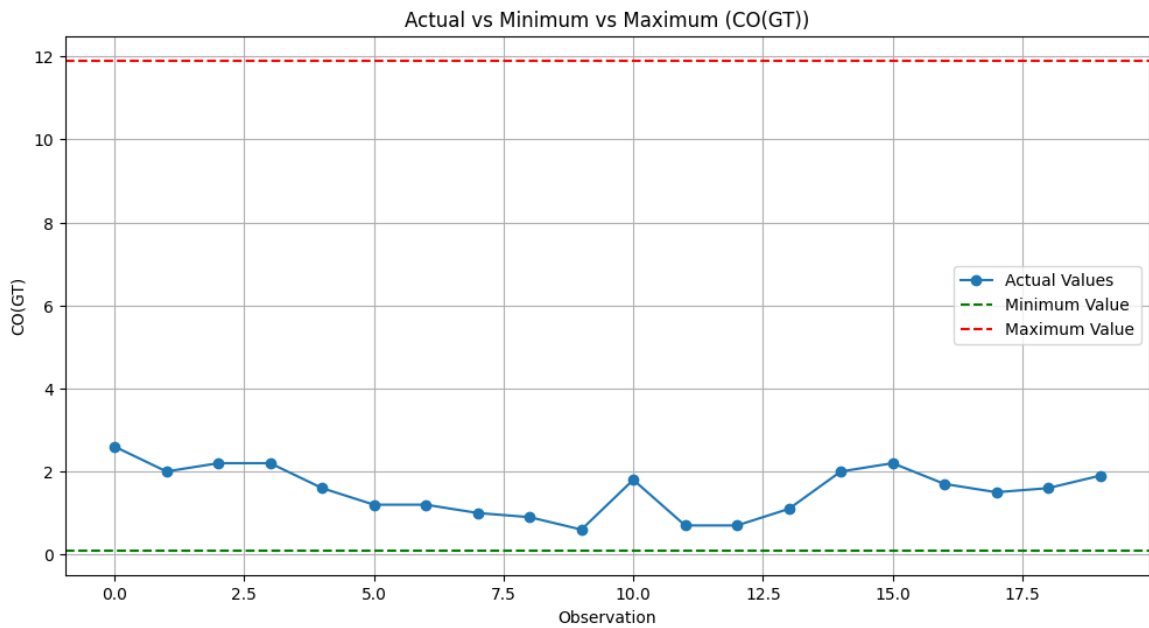
Histogram of Numerical Features



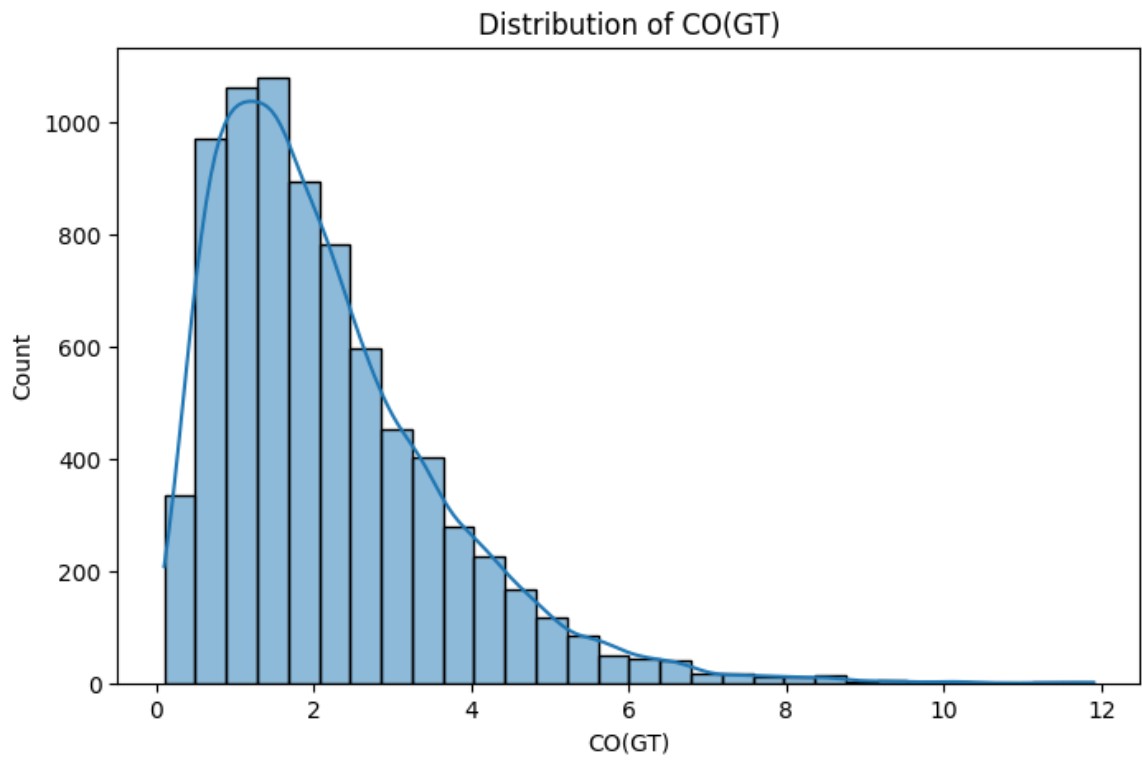
In [11]:



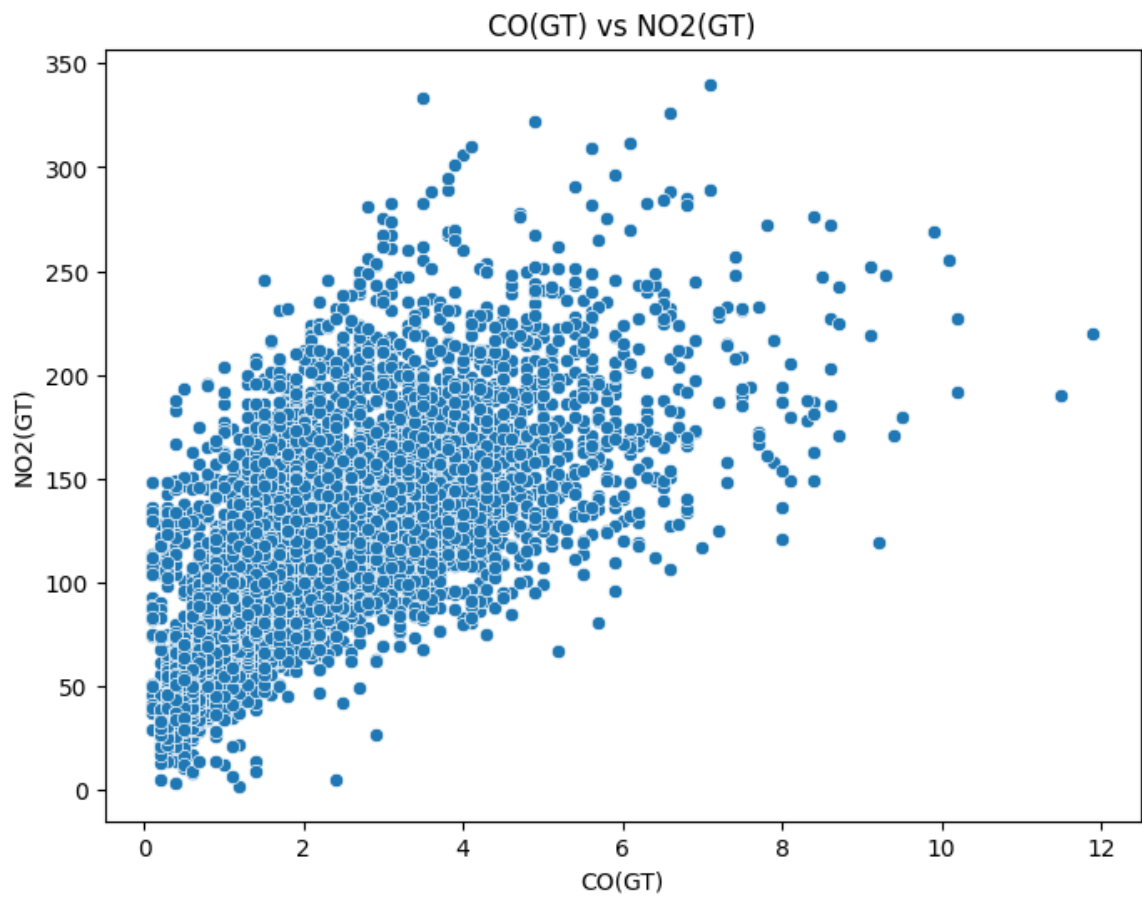
In [12]:



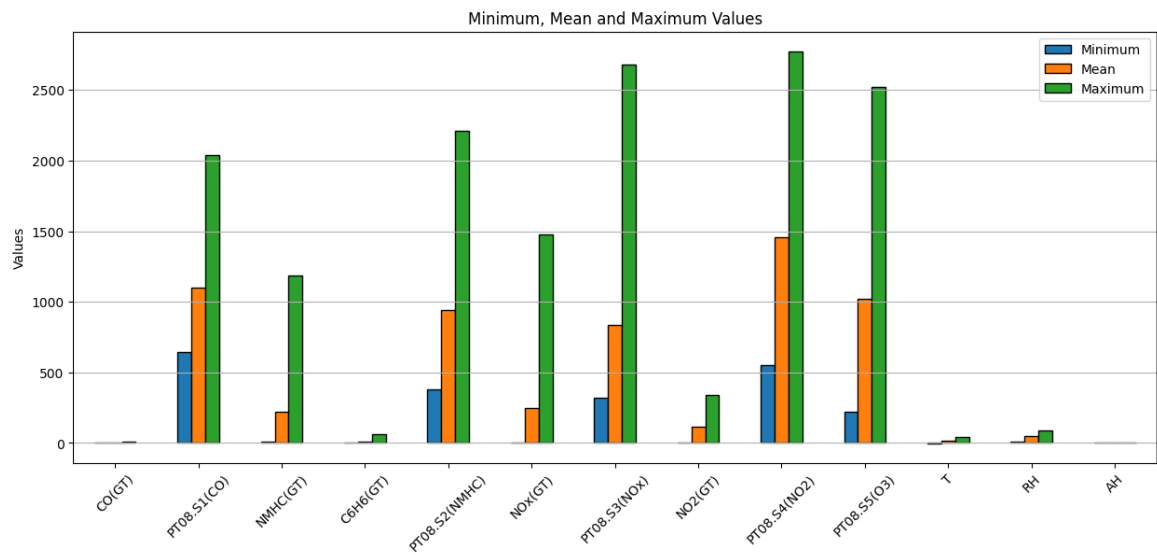
In [18]:



In [19]:



In [20]:



In []: